

Practical - 1

Practical Title: Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Problem Definition: Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.

A. Theory

1. What is URL and parts of URL?

Answer: Write your A URL (Uniform Resource Locator) is a unique address used to locate and access resources on the internet. A URL is composed of multiple parts, each defining how and where a resource can be accessed.

<https://www.geeksforgeeks.org/computer-networks/components-of-a-url/>

2. How does HTML file process in web browser?

Answer: A web browser processes an HTML file through a structured pipeline called the Critical Rendering Path (CRP), transforming raw code into the interactive visual page you see on your screen.

<https://pavankumar-patrani.medium.com/the-process-behind-browser-loading-a-webpage-9fd80792ea94>

3. How will page navigation flow managed among all HTML pages?

Answer: Page navigation flow across HTML files is managed using anchor tags (<a>) with relative or absolute file paths inside semantic <nav> menus, ensuring a consistent user journey.

<youtube.com/watch?v=jHNIFMOW6es&t=1>

4. How will GitHub commits be maintained after each practical?

Answer: After each practical, GitHub commits should be maintained by staging your completed work, writing a clear and descriptive commit message detailing what was done, and pushing the changes to your remote repository to secure a permanent history of your progress.

<https://www.theserverside.com/video/Follow-these-git-commit-message-guidelines>

B. Implementation/ Output/Result

Implementation of Wireframe

The **wireframe** of the **StudentHub Portal** was designed to represent the basic layout and structure of the website before development. It defines the placement of important elements such as the header, navigation menu, content sections, and footer.

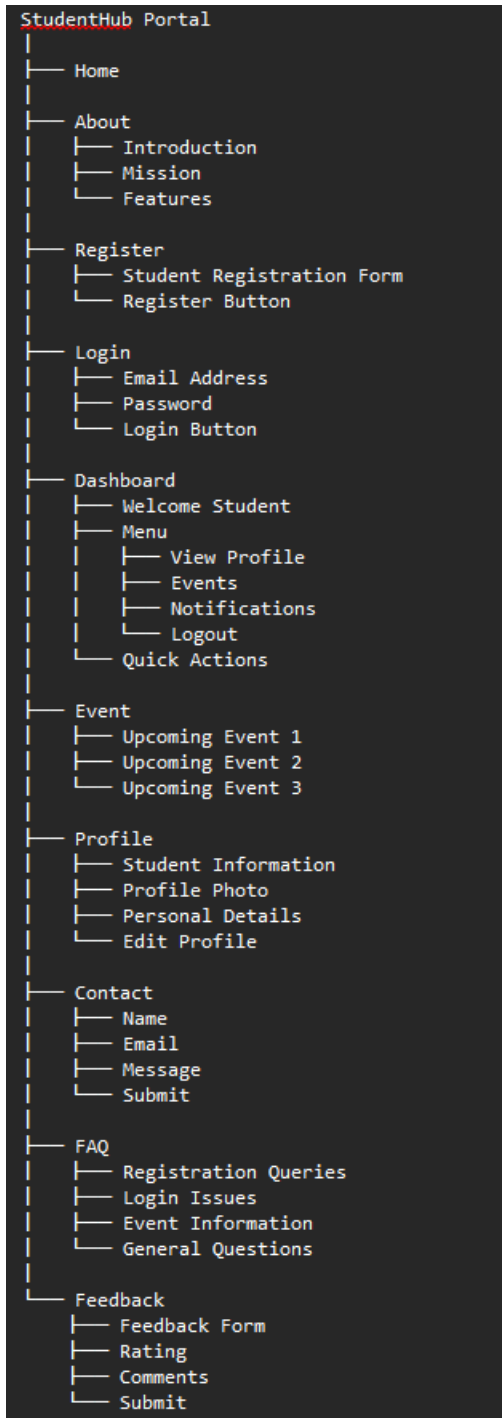
C. Evidence of Work (Screenshots)

The wireframe illustrates the StudentHub Portal interface. It features a top navigation bar with the following links: HOME, ABOUT, REGISTER, LOGIN, DASHBOARD, EVENT, PROFILE, CONTACT, FAQ, and FEEDBACK. The main content area is divided into several sections: 1. STUDENT REGISTRATION: A section with a REGISTER button. 2. STUDENT LOGIN: A section with input fields for EMAIL ADDRESS and PASSWORD, and a LOGIN button. 3. WELCOME STUDENT: A section with a MENU button. 4. UPCOMING EVENTS: A section with three placeholder boxes for event listings. 5. STUDENT PROFILE: A section with a large placeholder box for a student's profile information. 6. CONTACT US: A section with three horizontal input fields for contact information. 7. FREQUENTLY ASKED QUESTION: A section with three horizontal input fields for frequently asked questions. 8. FEEDBACK FORM: A section at the bottom for a feedback form.

Description:

The **StudentHub Portal** is a web-based application designed to provide students with a centralized platform for accessing academic and campus-related services. The portal offers a simple and user-friendly interface with multiple sections that help students

register, log in, view events, manage their profiles, contact the administration, access FAQs, and submit feedback.



Description:

A **sitemap** is a hierarchical diagram or list that shows the structure and organization of a website. It illustrates how all the web pages are connected and helps developers, designers, and users understand the website's navigation.

D. GitHub Repository Link:

<https://github.com/vatsal-1950/StudentHUB.git>

E. Learning Outcomes:

Students will be able to analyze requirements and prepare a structured project foundation before implementation.

Faculty Remarks: